
MAX School on Advanced Materials and Molecular Modelling
with QUANTUM ESPRESSO

QE-2021: Hands-on session – Day-7

(Magnetism in transition metals)

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Topics of Day-7 hands-on session:

1. Comparison of magnetic and non-magnetic solution for Fe (bcc)
([exercise1.ferromagnetic_vs_nonmagnetic/](#))
2. Dependence of exchange splitting on the lattice constant.
([exercise2.volume_dependence](#))
3. Comparison of $3d$ occupations in transition metals (Fe, Co, Ni) ([exercise3](#))
4. Plotting bands of magnetic material collinear and non collinear case.
([exercise4.bands](#))

How to run calculations remotely on the “hpc” HPC cluster

Several utility commands have been implemented specially for the QE-2021 school to aid at submitting jobs to HPC cluster(s). These are:

- **remote_mpirun** – this is like **mpirun**, but it automatically submits the calculation to a queuing system on the “hpc” HPC system. For example, a **pw.x** calculation can be submitted as:

- **remote_mpirun pw.x -in pw.file.in**

where **pw.file.in** is the name of the **pw.x** input file. **BEWARE:** stdin/stdout redirection does not work for **remote_mpirun**, hence you must use **-in** (or **-inp**) option (i.e., do not use “<” redirection operator). You do not need to specify the number of processors, because the default is set to **-np 20**.

A different number of processors, say 8, can be requested as:

- **NPROC=8 remote_mpirun pw.x -in pw.file.in**

- **remote_pwtk** – this automatically submits the PWTK script to queuing system on the “hpc” HPC system. Example:

- **remote_pwtk script.pwtk**

where **script.pwtk** is the name of the PWTK script.

- **remote_sbatch** – automatically submits the Unix-shell script to queuing system on the “hpc” HPC system. Example:

- `remote_sbatch script.sh`

where `script.sh` is the name of the Unix-shell script. When you run in the HPC cluster use increase the number of pools change the line:

```
mpirun -np $NPROC $PW < fe.scf.$latt.in -nk 2 > fe.scf.$latt.out
```

in

```
mpirun -np $NPROC $PW < fe.scf.$latt.in -nk 10 > fe.scf.$latt.out
```

- **remote_squeue** – this checks the status of the job in the queuing system on the “hpc” HPC system. Example:

- `remote_squeue`

JOBID	PARTITION	NAME	USER	ST	TIME	NODES	NODELIST(REASON)
26098354	all	submit.n	qe001	R	0:27	1	wn150

Where status (ST) R stands for running and PD for pending.

- **hpc** – this makes `ssh` to “hpc” HPC login node, such that the user will be located in the same directory as used locally
- **rsync_to_hpc** – copies specified files to the “hpc” cluster to the same directory as is currently used locally. Example:

- `rsync_to_hpc '*.in'`

This will copy all `*.in` files from local directory to the same directory on the “hpc” cluster, whereas:

- `rsync_to_hpc .`

will copy all the files in the current directory to the same directory on the “hpc” cluster.

- **`rsync_from_hpc`** – download the specified file from the “hpc” cluster from the same directory as is currently used locally. Example:

- `rsync_from_hpc '*.out'`

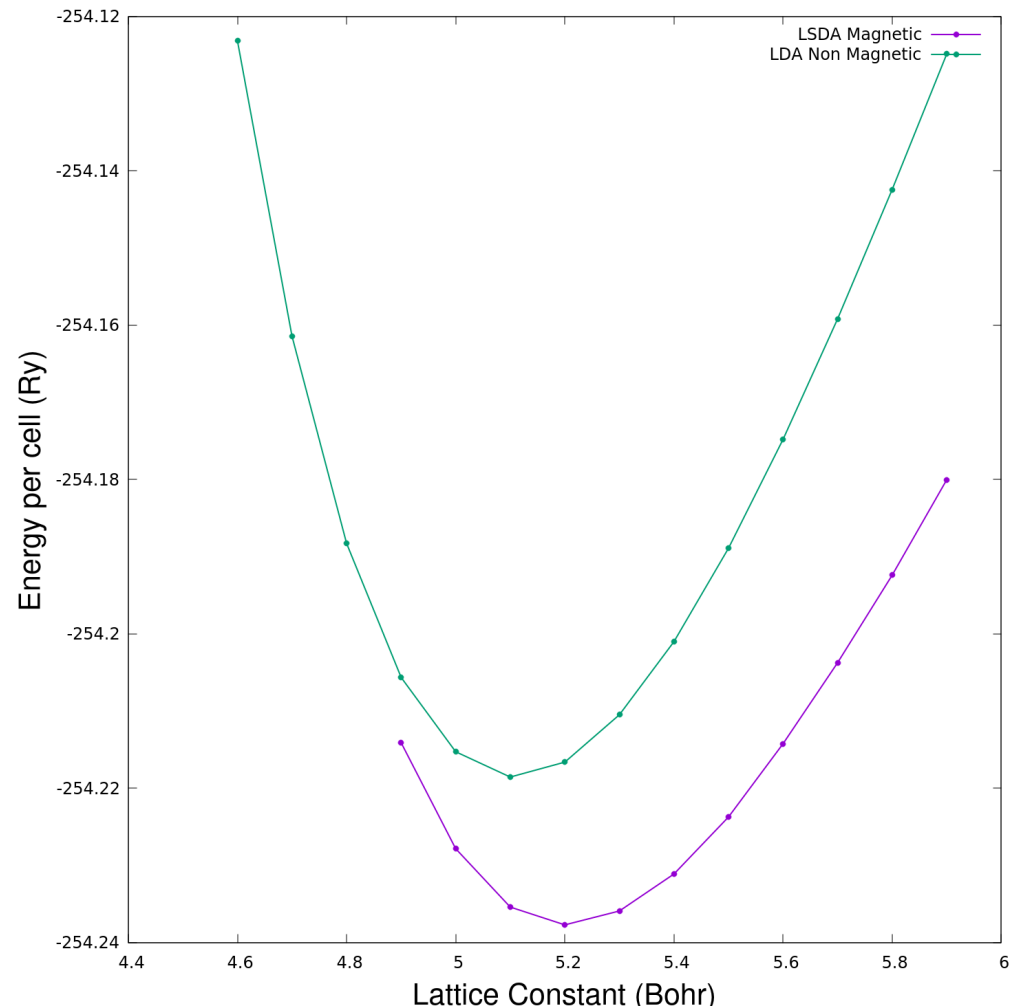
This will copy all `*.out` files from the “hpc” cluster to the local host (virtual machine).

Exercise 1: Task 1

- Compare the magnetic and the non-magnetic solution for Fe (bcc)
- **Find the optimized lattice constant for non-magnetic Fe (bcc):**
 1. go to `exercise1.ferromagnetic_vs_nonmagnetic/non_magnetic` :
 2. Use the scripts `job.sh` or `job.pwtk` to scan the lattice constants from 4.6 Bohr to 5.9 Bohr.
 3. use `ev.x` to fit the data collected in `energies.dat` with the Murnaghan EOS and find the optimized lattice constant.
- **Find the optimized lattice constant for magnetic Fe (bcc)**
 1. go to `/Day-7/exercise1.ferromagnetic_vs_nonmagnetic/magnetic`
 2. repeat steps 2 and 3 as for the non-magnetic case.
- **Plot and compare energies and lattice constants**
 - go to `/Day-7/exercise1.ferromagnetic_vs_nonmagnetic/`
 - use the gnuplot script `compare.gp` to plot your results.

Compare energies and lattice constants.

- The magnetic solution has a lower total energy per cell.
- The magnetic solution has a larger lattice constant than the non-magnetic.
- LSDA underestimate lattice constant ~ 5.2 Bohr vs. ~ 5.4 Bohr of the experiment.



Remarks on input

How to select the magnetic solution:

```
&system  
 ibrav=3,  
 celldm(1)=$latt,  
 nat= 1,  
 ntyp=1,  
 nspin=2,  
 starting_magnetization(1)=0.3,  
 ecutwfc=70.0, ecutrho=850.0,  
 occupations='smearing',  
 smearing='marzari-vanderbilt',  
 degauss=0.02  
/
```

- `nspin=2` Specify the spin polarized calculation.
- To allow a magnetic solution we have also break the symmetry between the *up* and *down* channel:
 - `starting_magnetization(1)=0.3`
Builds the starting potential assuming an initial magnetization of Fe of 0.3×16 .

Remarks on input

Magnetic system in the non-collinear case

```
&system
 ibrav=3
celldm(1)=5.4,
nat=1,ntyp=1,
noncolin=.true.,
starting_magnetization=0.3,
angle1(1)=0.0,
angle2(1)=0.0,
ecutwfc=70.0,
ecutrho=850.0,
occupations='smearing',
smearing='marzari-vanderbilt',
degauss=0.02
/
```

- non-collinear case is selected specifying `noncolin=.true.`
- `starting_magnetization` must be specified for at least one species otherwise the program will perform a non-magnetic calculation.
- in the non-collinear case magnetization is a vector, it is possible to specify the direction of the starting magnetization for each atomic type:
 - `angle1` is the angle with the z axis (in degrees);
 - `angle2` is the azimuthal angle (in degrees)

How the magnetization is reported in output

- In the spin-polarized case the total magnetization is reported at each SCF step and at the end of the calculation:

total magnetization	=	2.22 Bohr mag/cell
absolute magnetization	=	2.28 Bohr mag/cell

- In the magnetic non-collinear case:

total magnetization	=	1.38	0.00	1.38 Bohr mag/cell
absolute magnetization	=	1.99 Bohr mag/cell		

- An estimate of the magnetization at each ionic site is also printed at the end of the calculation

Magnetic moment per site (integrated on atomic sphere of radius R)				
atom	1 (R=0.357)	charge=	14.4267	magn= 2.2287

- In the magnetic non-collinear case:

```
atom number      1 relative position :      0.0000      0.0000      0.0000
charge :      14.332507 (integrated on a sphere of radius 0.357)
magnetization :           1.374808      -0.000000      1.374808
magnetization/charge:      0.095922      -0.000000      0.095922
polar coord.: r, theta, phi [deg] :      1.944273      45.000000      -0.000000
```

Input of dos.x and projwfc.x

```
&dos
  prefix='fe',
  bz_sum='tetrahedra_opt'
  Emin=5.0
  Emax=25.0
  DeltaE=0.05
  fildos='fe.dos'
/
```

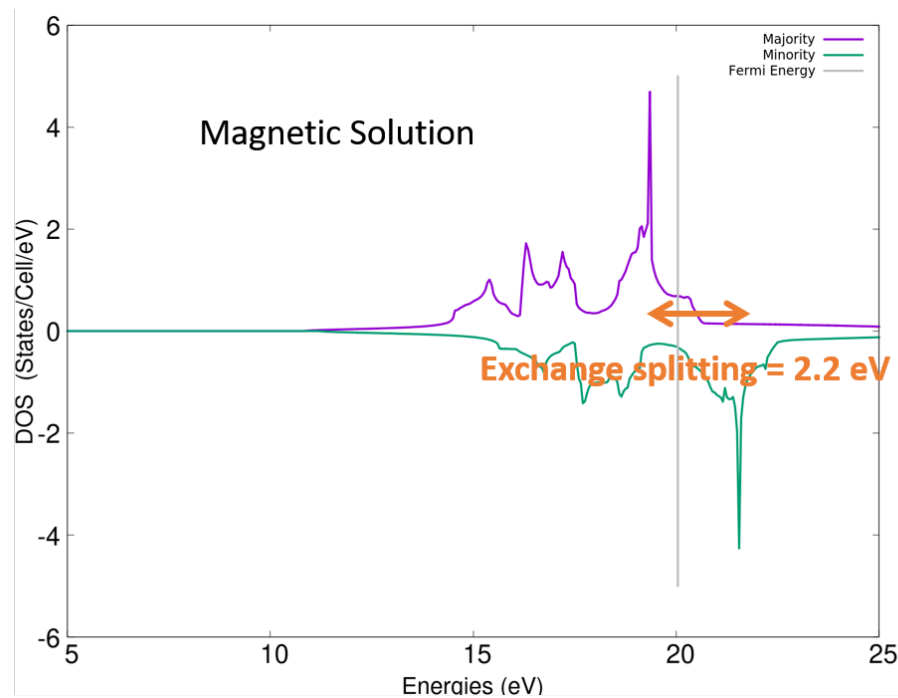
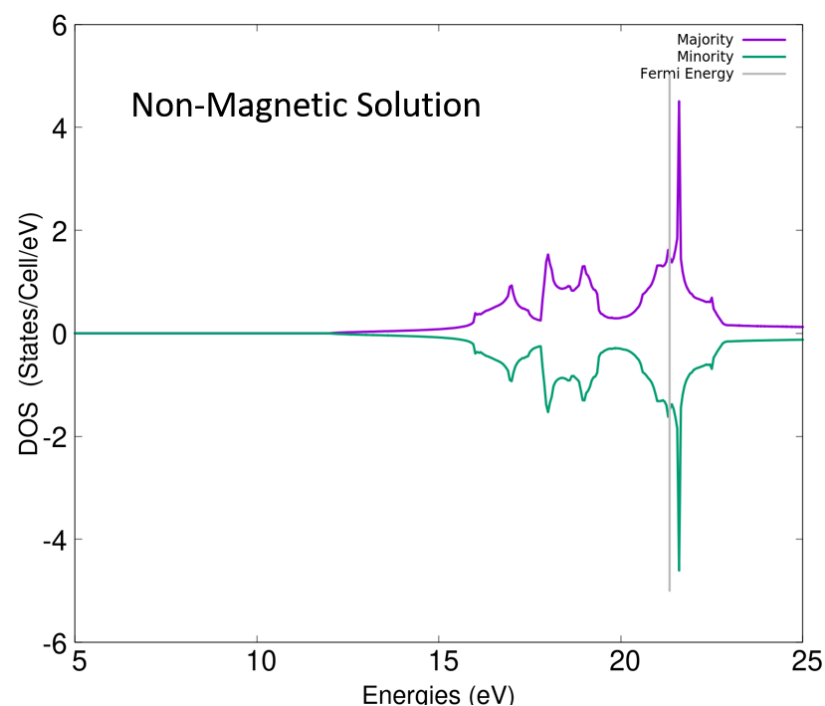
- **bz_sum** selects the integration method.
- **Emin**, **Emax**, and **DeltaE** define the range and definition of the DOS;
- to select the range first use default values, then select the right window

```
&projwfc
!outdir='./tempdir/',
prefix='fe'
Emin=5.0, Emax=25.0, DeltaE=0.05
/
```

Exercise 1: Task 2

Using the optimized lattice constants obtained in task 1 we analyze the electronic structures of the two solutions.

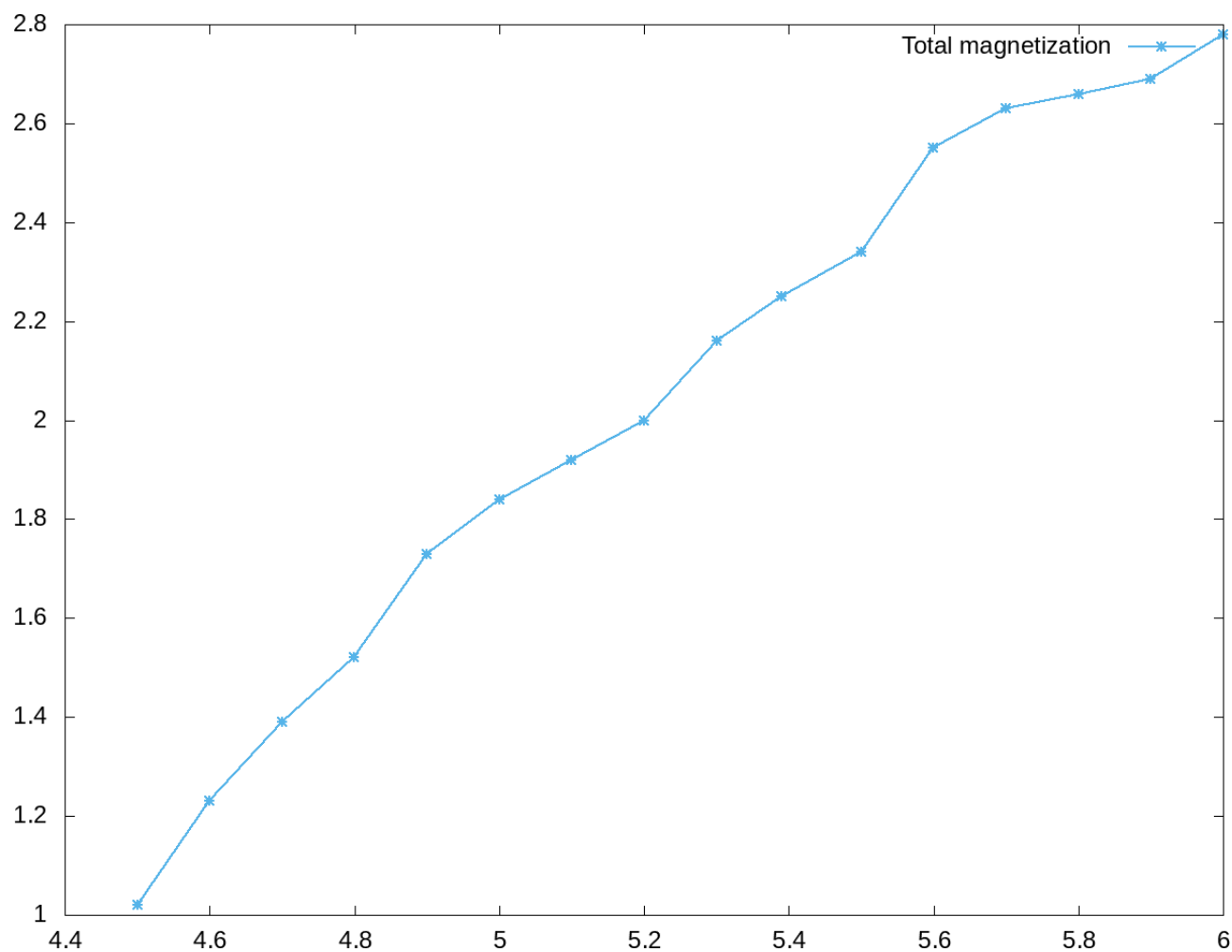
1. Make the SCF and non-SCF computations for boths solutions.
2. Compute the DOS for both solutions
3. Compute the pDOS for the magnetic solution



Exercise 2: Magnetization vs. volume

- The aim of this exercise is to see how the magnetization and the exchange splitting behaves when the lattice constant decreases
- Here we use the PBE functional that gives a more realistic lattice constant than LDA.
- The exercise proceeds along the following steps:
 1. Run a script for determining the optimized lattice constant
 2. Use `ev.x` to estimate the optimized lattice constant fitting the energies to the Murnaghan EOS.
 3. Use the data in the file `magnetization.dat` to see how the magnetization changes when we change the lattice constant.
 4. **Assignment** Compute the electronic structure for 3 different values of lattice constant and estimate the the exchange splitting as we did in exercise 1.

Magnetization vs. alat



Exercise 3. Comparing magnetization and exchange splitting in transition metals

In this exercise we compare the magnetic properties of three different transition metals Fe(bcc), Co(hcp), and Ni(fcc)

- With the usual procedure we compute the electronic structure for the 3 transition metals.
- use `projwf.x` to analyze the contribution to the magnetization coming from the different orbitals.

```
Atom #    1: total charge = 17.0232, s = 2.4486, p = 6.1080, d = 8.4666,  
spin up      = 8.8345, s = 1.2198,  
spin up      = 8.8345, p = 3.0211, pz= 1.0070, px= 1.0070, py= 1.0070,  
spin up      = 8.8345, d = 4.5935, dz2= 0.9308, dxz= 0.9106, dyz= 0.9106,  
              dx2-y2= 0.9308, dxy= 0.9106,  
spin down    = 8.1887, s = 1.2288,  
spin down    = 8.1887, p = 3.0869, pz= 1.0290, px= 1.0290, py= 1.0290,  
spin down    = 8.1887, d = 3.8731, dz2= 0.8427, dxz= 0.7292, dyz= 0.7292,
```

dx²-y²= 0.8427, dxy= 0.7292,
polarization = 0.6457, s = -0.0089, p = -0.0658, d = 0.7204,
Spilling Parameter: 0.0543

Format of projwfc.x output

- The total DOS and the sum of projected DOS are written to file `<filpdos>.pdos_tot`.
 - The format for the collinear, spin-unpolarized case and the non-collinear, spin-orbit case is:

```
E DOS(E) PDOS(E)
...
```

- The format for the collinear, spin-polarized case is:

```
E DOSup(E) DOSdw(E) PDOSup(E) PDOSdw(E)
...
```

- The format for the non-collinear, non spin-orbit case is:

```
E DOS(E) PDOSup(E) PDOSdw(E)
...
```

Format of the projection files

In the collinear case and the non-collinear, non spin-orbit case projected DOS are written to file `<filpdos>.pdos_atm#N(X)_wfc#M(l)`, where N = atom number , X = atom symbol, M = wfc number, l=s,p,d,f (one file per atomic wavefunction found in the pseudopotential file)

- The format for the collinear, spin-unpolarized case is:

E LDOS(E) PDOS_1(E) ... PDOS_2l+1(E)
...

where

- $LDOS = \sum_{m=1,2l+1} PDOS_m(E)$
 - $PDOS_m(E)$ = projected DOS on atomic wfc with component m
- The format for the collinear, spin-polarized case and the non-collinear, non spin-orbit case is as above with two components for both LDOS(E) and PDOS_m(E)

Orbital order

- for $l=1$:

1. p_z ($m=0$)
2. p_x (real combination of $m=+/-1$ with cosine)
3. p_y (real combination of $m=+/-1$ with sine)

- for $l=2$:

1. d_{z^2} ($m=0$)
2. d_{zx} (real combination of $m=+/-1$ with cosine)
3. d_{zy} (real combination of $m=+/-1$ with sine)
4. $d_{x^2-y^2}$ (real combination of $m=+/-2$ with cosine)
5. d_{xy} (real combination of $m=+/-2$ with sine)

Exercise 4: plotting bands

- this exercise provides an example on how to plot the bands of a magnetic system.
- Spin-polarized collinear case:
 - • `cd Ni_collinear`
 - SCF calculation for Ni (bcc): • `pw.x < ni.scf.in > ni.scf.out`
 - non-SCF calculation:
 - * `calculation='bands'`
 - * set the list of K-points

```
K_POINTS crystal_b
6
0.000 0.000 0.000 20 !gamma
0.500 0.500 0.500 10 !L
0.500 0.250 0.750 10 !W
0.500 0.000 0.500 10 !X
0.000 0.000 0.000 20 !gamma
0.375 0.375 0.750 1 !K
```

- we specify the high-symmetry points and number of points of each segment
- `crystal_b` specify the format for K-points

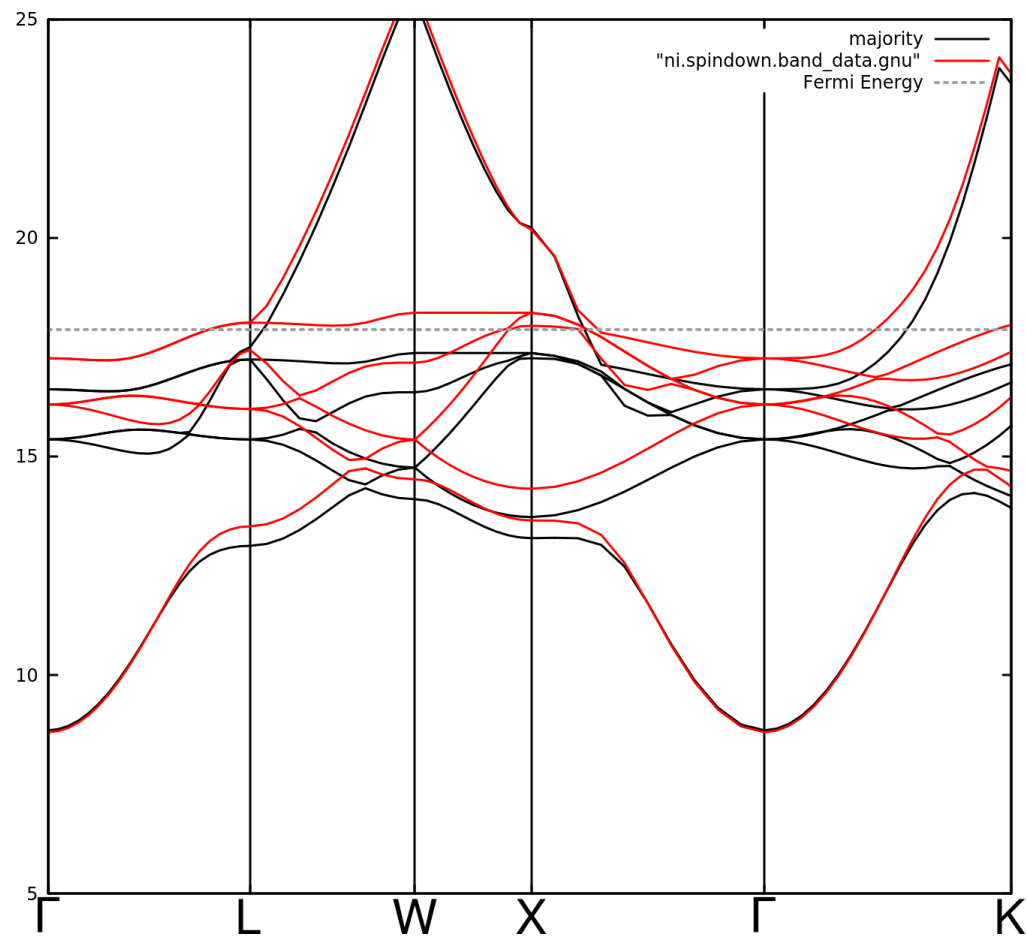
- use `bands.x` for preparing the band plot and analyze symmetry, first for up component then for down component

&BANDS

```
outdir='./tempdir/',  
prefix='ni',  
filband='ni.spinup.band_data',  
spin_component = 1,  
/
```

- * `spin_componet` selects the spin component
- * `filband` selects the prefix where the bands are saved

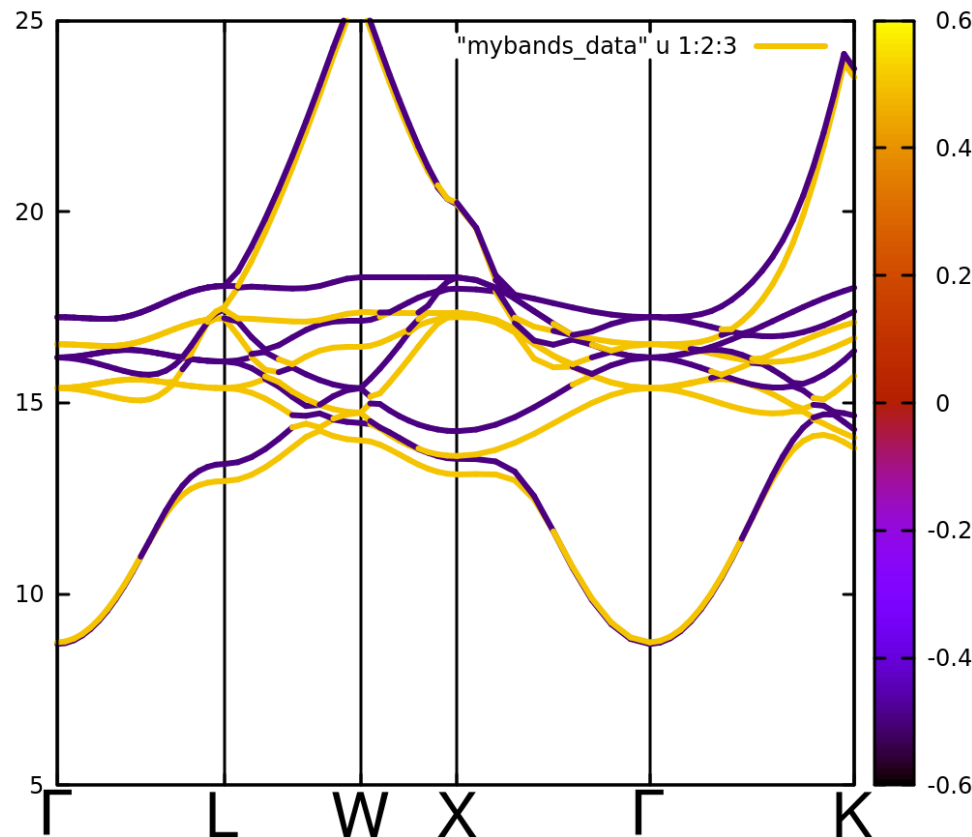
- the output of `bands.x` provides the length of the k-points path and the position of the high-symmetry points
- use the gnuplot script to plot the bands



Non-collinear case

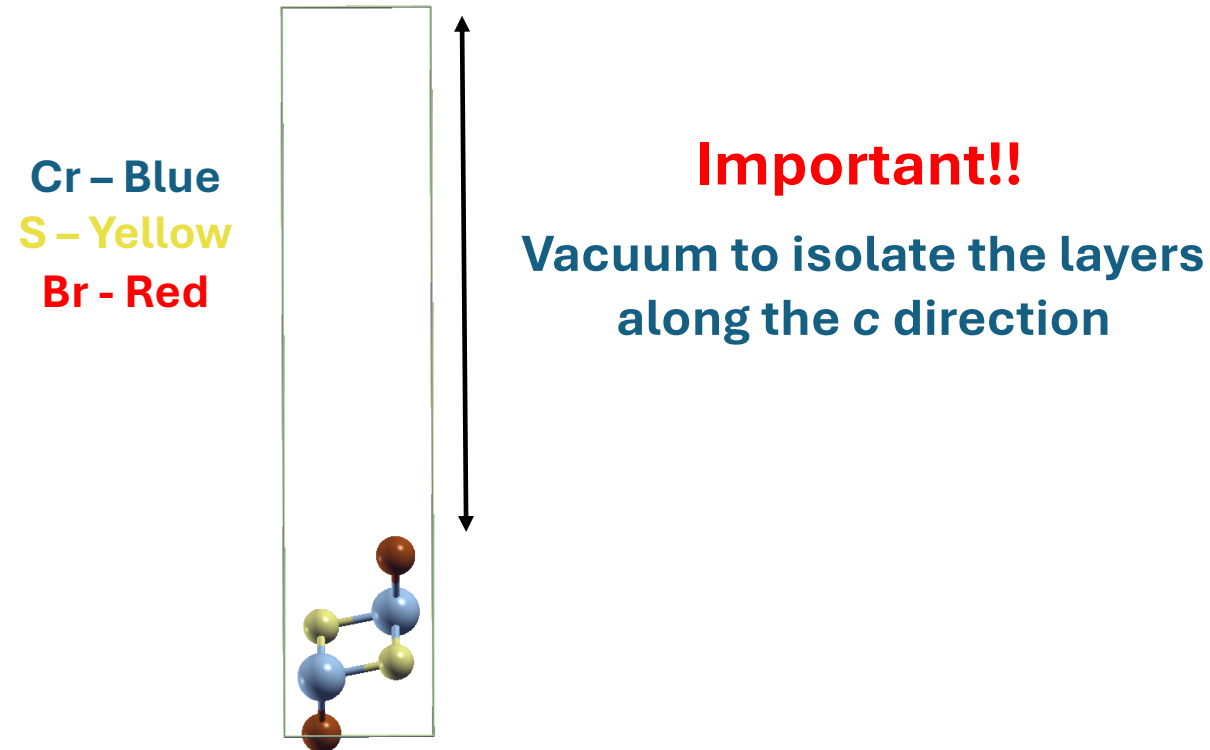
- Plot the Ni bands in the noncollinear case:
 - • `cd Ni_noncollinear`
 - run the collinear scf calculation for Ni in this directory:
 - `pw.x < ni.scf.in > ni.scf.out`
 - run the non-collinear nscf calculation for the bands
 - * `spin=2` has been replaced with `noncollinear=.true.`
 - `pw.x < ni.bands.in > ni.bands.out`
 - run 'bands.x' for the noncollinear case:
 - `spin_component` has been removed and we add `lsigma(3)=.true.` that instructs the program to compute the expectation value for the z component of the spin operator for each eigenfunction and save all values in the file `ni.noncolin.data.3`. All values in this case are either 1/2 or -1/2 as expected.
 - the program `plot_noncolin_bands.f90` reads these values and writes them together with the band structure in the file `my_bands.data`.
 - * compile the program:
 - `gfortran -o mino.x plot_noncolin_bands.f90`

- * copy `ni.noncollin.data.3` to `ni.noncollin.data.s`
- * run the program
 - `./mino.x ni.noncollin.data`
- * use gnuplot and the script `bands_noncollin.gp` to plot the bands in this case.
- * start gnuplot and type the command:
 - `gnuplot> load "bands_noncollin.gp"`



Exercise 5: plotting bands for a 2D material

- This exercise provides an example on how to plot the bands of CrSBr, a ferromagnetic and semiconducting 2D material.
- To capture the 2D dimensionality, we should include vacuum along the c direction to avoid interactions between the periodic images



Remarks on input

calculation='scf'

```
&system  
ibrav=8,  
celldm(1)=6.805597590,  
celldm(2)=1.337896944,  
celldm(3)=6.544944737,
```

A large **celldm(3)** is used to model an isolated CrSBr monolayer. Usually, around 18 Å of vacuum is enough to isolate layers

calculation='bands'

```
K_POINTS crystal_b  
5  
0.000 0.000 0.000 10 ! Gamma  
0.500 0.000 0.000 10 ! X  
0.500 0.500 0.000 10 ! S  
0.000 0.500 0.000 10 ! Y  
0.000 0.000 0.000 1 ! Gamma
```

in_plane: Γ -X-S-Y- Γ
Band structure within the 2D plane.

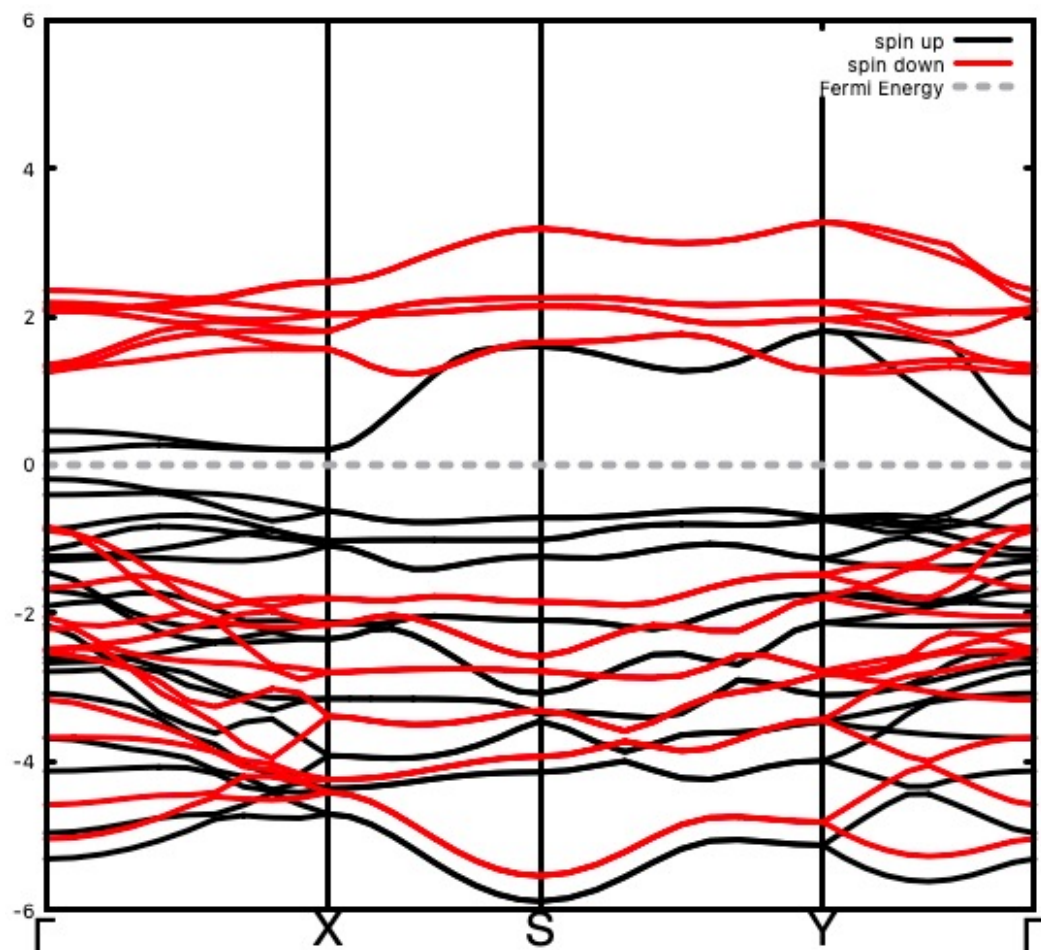
out_of_plane: Γ -A
Check dispersion along c; bands
should be nearly flat given that
layers do not interact.

```
K_POINTS crystal_b  
6  
0.000 0.000 0.000 10 ! Gamma  
0.500 0.000 0.000 10 ! X  
0.500 0.500 0.000 10 ! S  
0.000 0.500 0.000 10 ! Y  
0.000 0.000 0.000 10 ! Gamma  
0.000 0.000 0.500 1 ! A
```

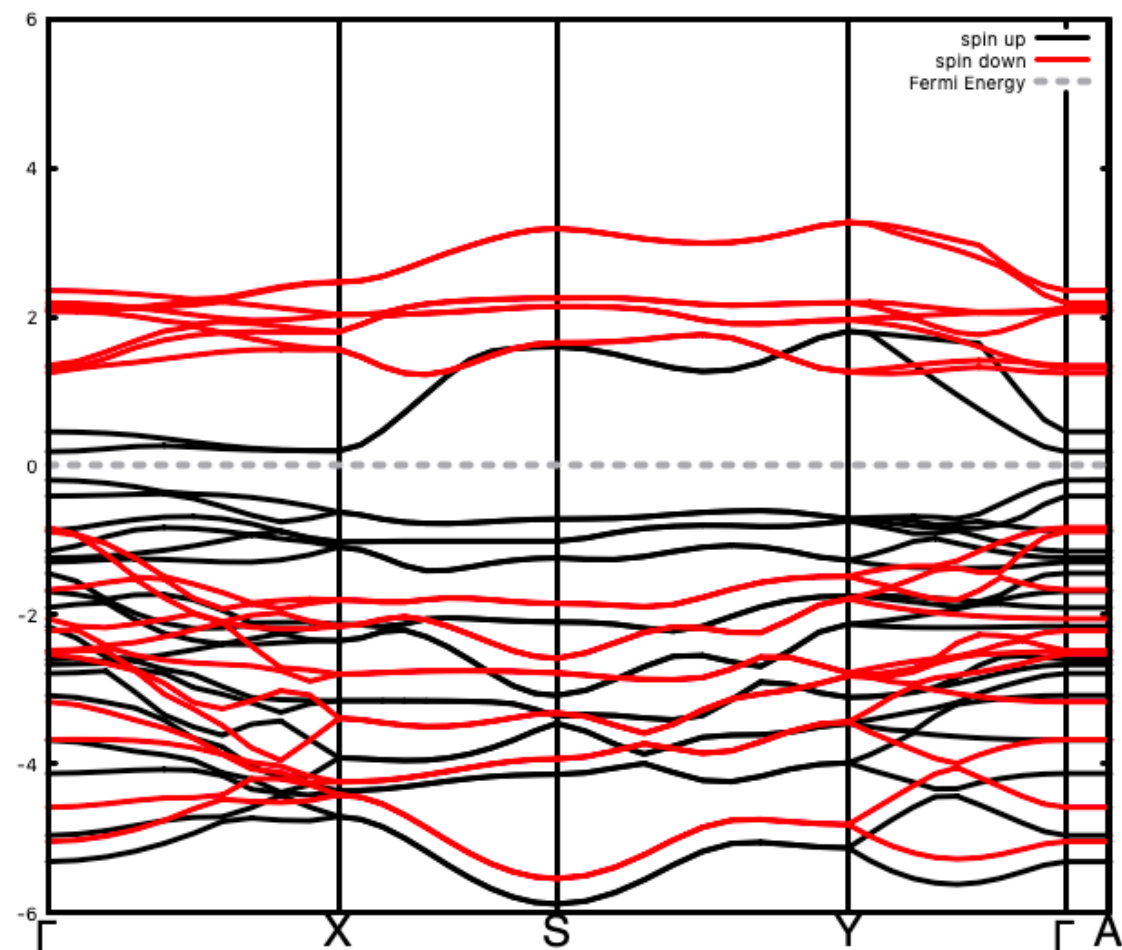
cd exercise5.bands.2Dmaterial/in_plane/

cd exercise5.bands.2Dmaterial/bands_out_of_plane/

cd exercise5.bands.2Dmaterial/in_plane/



cd exercise5.bands.2Dmaterial/bands_out_of_plane/



THE END